

BALWANT SINGH BISHT

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Data Scientist with industry and research experience in machine learning, AI applications, and data engineering. Experienced in developing conversational AI systems, ETL pipelines, and privacy-preserving machine learning solutions. M.Tech in Computer Science (Data Science) from the Indian Statistical Institute.

Technical Skills

Programming skills — Python • SQL

Data Analysis — Pandas • Numpy • Matplotlib

Machine Learning — PyTorch • Scikit-learn

Data Engineering — Airflow • BigQuery • REST APIs • ETL

AI — Deep Learning • Federated Learning • NLP • Conversational AI • Prompt Engineering • Azure AI Services

Developer Tools — Git • GitHub • Jupyter Notebook • Google Colab • VS Code

Experience

Associate Data Scientist — Zee Entertainment Enterprises Limited Jul 2023 — Mar 2024

- Developed a conversational business intelligence system enabling natural-language querying of enterprise data by integrating Azure Speech-to-Text, GPT-4, SQL generation, and Google BigQuery.
- Built automated ETL pipelines using Python, Apache Airflow, REST APIs, and BigQuery for ingesting, transforming, and storing Facebook and Instagram analytics data.
- Implemented SQL-based data transformation and reporting workflows supporting downstream analytics and business intelligence.

Junior Research Fellow — Netaji Subhash University of Technology Oct 2024 — Dec 2025

- Conducted research on privacy-preserving federated learning using model pruning and secure gradient masking.
- Implemented and evaluated a federated learning framework based on the Lottery Ticket Hypothesis and analyzed communication efficiency, privacy, and model performance.
- Contributed to a research manuscript describing the proposed framework and experimental findings.

Mathematics Teacher — Anand School For Excellence Hansi Mar 2019 — Sep 2020

- Taught Mathematics to students from Classes V–XII and developed online learning content during the COVID-19 pandemic.
- Mentored students for the National Science Exhibition.

Education

M. Tech in Computer Science (Data Science) – **Indian Statistical Institute**, Kolkata Oct 2021 — Jun 2023

M. Sc. in Physical Sciences – **S. N. Bose National Centre For Basic Sciences**, Kolkata Aug 2014 — Jul 2016

B. Sc. in Physics (Hons.) – **Kirorimal College, University of Delhi**, New Delhi Jul 2011 — Jun 2014

Featured Project

[Sentiment Analysis on Code-Mix Data](#) — M.Tech Dissertation

Jan 2023 — Jun 2023

- Fine-tuned the XLM-RoBERTa transformer model for multilingual sentiment analysis on code-mixed Indian language datasets.
- Proposed a novel approach to mitigate data scarcity by combining code-mixed datasets from linguistically related Indian languages sharing the Devanagari script.
- Evaluated the proposed approach using standard classification metrics and demonstrated improved model performance over individual-language training.

Project Repository

Publication

[Pruned FL with Secure Gradient Masking for Privacy-Preserving and Efficient Model Training](#)

Co-author, submitted.

Achievements

[UGC-NET \(Physical Sciences\)](#) — All India Rank 31

2018

[Joint Entrance Screening Test \(JEST\) - Physics](#) — All India Rank 96

2014